ASCII MASTER FLOW — PANEL 1/12: Transport to trade chain

```
MINE OR FARM -> production point with dispersed origins
ROADHEAD -> flexible collection over short hauls
RAIL OR WATERWAY CORRIDOR -> bulk trunk haul over long distance
PORT OR AIRPORT -> break of bulk and interface with global routes
WORLD MARKET -> exchange completed through connectivity, cost and time
MUST REMEMBER: Transport geography explains networks through nodes, links, accessibility,
  connectivity, density, hierarchy, break-of-bulk, modal cost and hinterland; trade adds
  comparative advantage, corridors, ports, logistics and chokepoints, while India's space
  programme links launch, satellite, application and governance institutions.
```

ASCII MASTER FLOW — PANEL 2/12: Modal cost structure comparison

ROAD -> low terminal cost, high per-km cost; short hauls and door-to-door delivery
RAIL -> higher terminal cost, lower per-km cost; medium to long bulk hauls
WATER -> highest terminal cost, lowest per-km cost; long bulky low-value cargo
AIR -> costly throughout; justified by high value-to-weight, urgency or inaccessibility
PIPELINE -> very high fixed cost, very low operating cost; continuous liquid or gas flow

ASCII MASTER FLOW — PANEL 3/12: Modal choice decision rule

```
IDENTIFY CARGO -> bulk, value density, urgency and perishability
IF haul is short -> terminal cost dominates, road usually wins
IF haul exceeds the road-rail break-even -> rail or water recovers its terminal cost
IF value-to-weight is very high or access is difficult -> air becomes defensible
ALWAYS -> treat freight as an intermodal chain, not a single modal verdict
```

ASCII MASTER FLOW — PANEL 4/12: Network vocabulary rail

HINTERLAND -> inland area served by a port, city or route
NODE -> junction or terminal where flows converge
CORRIDOR -> linear axis of intense transport and economic activity
BREAK OF BULK -> port, railhead or dry port where cargo changes mode
HUB AND SPOKE -> one major hub redistributes flows to smaller nodes

ASCII MASTER FLOW — PANEL 5/12: Port site versus situation

```
SITE -> depth, shelter, tidal range, foreshore, expansion land, siltation risk
SITUATION -> hinterland productivity, inland connectivity, position on shipping lanes
CORRECTABILITY -> site can be dredged and engineered; situation cannot
GROWTH RULE -> situation normally decides which ports grow
POLICY -> freight corridors and inland container depots are port instruments
```

ASCII MASTER FLOW — PANEL 6/12: Port type worked comparison

```
NATURAL HARBOUR -> excellent site; stays small if the hinterland is poor
ESTUARINE PORT -> good route situation, but siltation and draught limit the site
ARTIFICIAL DEEP-WATER PORT -> engineered site chosen for hinterland reach and rail links
SATELLITE CONTAINER TERMINAL -> deep draught and back-up land beside a congested parent
TRANSHIPMENT HUB -> situation is the route itself, so it lives without a local hinterland
```

ASCII MASTER FLOW — PANEL 7/12: Containerisation hierarchy effect

```
STANDARD CARGO UNIT -> handling time and cost collapse at every transfer
INTERMODAL TRANSFER BECOMES ROUTINE -> production fragments across countries
ADVANTAGE SHIFTS -> deep draught, crane productivity and inland rail connectivity
TRAFFIC CONCENTRATES -> fewer, larger hubs with feeder services to the rest
ANSWER USE -> a hierarchy argument that needs no throughput statistic
```

ASCII MASTER FLOW — PANEL 8/12: Chokepoint vulnerability rail

NARROW PASSAGE -> world shipping is funnelled through a small cross-section
DISRUPTION -> voyages lengthen and rerouting begins immediately
COST TRANSMISSION -> freight and insurance rise far beyond the affected region
POLICY RESPONSE -> chokepoint security, alternative routes and canal capacity
READING -> the passage is valuable as route control, not as a water body

ASCII MASTER FLOW — PANEL 9/12: Transport and development two-way fork

```
NEW LINK OPENS -> relative accessibility of every node changes, not only the two joined
IF junctions, local loading and complementary investment exist -> corridor growth follows
IF they are absent -> the tunnel effect carries benefit to the endpoints
IF local producers are weak -> backwash exposes them and speeds out-migration
VERDICT -> connectivity is necessary but not sufficient for regional development
```

ASCII MASTER FLOW — PANEL 10/12: India multimodal corridor ladder

```
BHARATMALA -> economic corridors, inter-corridors, border and port connectivity
DEDICATED FREIGHT CORRIDORS -> separate freight from passenger congestion on major axes
SAGARMALA -> port modernisation, port-led industrialisation and coastal connectivity
PM GATI SHAKTI -> network integration replacing mode-wise sectoral planning
TRADE MAP -> western container and petroleum gateway; eastern bulk and mineral coast
CLOSE DISTINCTION: Road/rail density differs from effective accessibility, port capacity from
  throughput, inland waterway declaration from operational traffic, trade balance from current
  account, ISRO from the Department of Space and NewSpace India Limited, and
  navigation/communication/earth-observation missions retain distinct functions.
```

ASCII MASTER FLOW — PANEL 11/12: Space as applied geography

```
EARTH OBSERVATION -> land use, crop condition, forests, water bodies, coasts, glaciers
METEOROLOGY -> cyclone tracking and monsoon monitoring feeding warning systems
NAVIGATION AND COMMUNICATION -> positioning, timing and remote-region connectivity
GIS INTEGRATION -> imagery plus terrain, infrastructure and socio-economic layers
OUTPUT -> site selection, corridor alignment, watershed planning, damage assessment
EVIDENCE LIMIT: Golden Quadrilateral, Dedicated Freight Corridors, Bharatmala, Sagarmala,
  major ports, National Waterway-1, Delhi–Mumbai corridor, Malacca/Hormuz/Suez and
  PSLV/GSLV/LVM3, INSAT/GSAT, IRS/EOS and NavIC form named map-institution anchors; project
  and mission claims require official date/status qualification.
```

ASCII MASTER FLOW — PANEL 12/12: Space institutions and the operational-orbit caution

```
ISRO -> national space architecture organised around application, not prestige
SRIHARIKOTA -> east-coast launch site with over-sea trajectory safety advantage
NSIL AND IN-SPACe -> commercialisation and non-governmental participation
NavIC -> indigenous navigation supporting transport timing and positioning
NVS-02 ON 29 JAN 2025 -> reached transfer orbit but not its intended NavIC slot
```